Enterprise Architecture Blueprint: Multi-Region Kubernetes Deployment

Infrastructure Engineering Reference Architecture | ISO/IEC 27001 Compliance Audit Verified | Status: Approved

1. Architecture Overview

This technical specification outlines the production cloud hosting blueprint across AWS and Google Cloud Platform (GCP). In full alignment with our Corporate Security Policy and annual Compliance Audit standards, the architecture implements Zero Trust Network Access (ZTNA), mutual TLS 1.3 (mTLS) service-to-service communication, and declarative GitOps deployments via ArgoCD.

Key Architectural Tenets: • Zero Trust Principles: Every service identity is validated via SPIFFE/SPIRE cryptographic tokens. • Immutable Infrastructure: Worker nodes are ephemeral and provisioned via Terraform modules. • End-to-End Encryption: All data in transit is encrypted with TLS 1.3; data at rest utilizes customer-managed KMS keys. • High Availability: Active-active multi-region clusters with automatic BGP Anycast routing failover. • Compliance Verification: Audited under ISO/IEC 27001 governance frameworks.

2. Cluster Topology & Network Segmentation

The production topology utilizes isolated Virtual Private Clouds (VPCs) connected through private Cloud Interconnect gateways:

VPC Subnet	CIDR Block	Purpose / Workloads	Egress Filtering Policy
dmz-ingress-us-east	10.100.10.0/24	Application Load Balancers & WAF	Restricted to Cloudflare CDN IPs
apps-k8s-us-east	10.100.20.0/22	Production Microservices & API Pods	Internal mTLS Gateway Only
db-storage-us-east	10.100.30.0/24	PostgreSQL HA & Redis Cache	Strict No-Public-Internet Route
mgmt-ci-cd-us-east	10.100.40.0/24	GitOps Runners & Telemetry Agents	Outbound HTTPS to Authorized Registry

3. Security Hardening Controls

• **Pod Security Standards**: Mandatory Restricted Profile enforcing read-only root filesystems and non-root UID 10001.

• **Network Policies**: Default-deny ingress and egress rules between Kubernetes namespaces.

• **Secrets Management**: Dynamic short-lived database credentials injected via HashiCorp Vault sidecars.

4. Operational Telemetry & Monitoring

Cluster health, request latencies, and resource consumption are monitored through a centralized Prometheus and Grafana stack. Synthetic canary probes execute every 30 seconds across global points of presence to guarantee 99.99% availability.

Authorized Internal Documentation Repository: https://wiki.internal.lan/cloud-architecture

Classification: BENIGN / SAFE (Corporate Security Policy & Compliance Audit Passed)